

Query Analysis Report Project: ULK Academic Activity and Resource Management System Phase 7: Query Processing Analysis

1. 1. Introduction This report analyzes three complex queries executed on the PostgreSQL database system. The analysis focuses on execution plans, join strategies, and computational overhead.

2. 2. Query 1: Join of Multiple Tables SQL:

```
SELECT s.full_name, s.student_code, e.grade
FROM Student s
JOIN Enrollment e ON s.id = e.student_id
JOIN Course c ON e.course_id = c.id
WHERE c.title = "Advanced Databases Part 3";
```

2.1. Execution Plan Analysis

- **Filter Execution:** PostgreSQL first performs a Sequential Scan on the Course table to find the record where title matches.
- **First Join (Hash Join):** The result from the Course filter is used as a hash table. The Enrollment table is scanned, and matches are identified.
- **Second Join (Nested Loop):** The filtered enrollments are then joined with the Student table.
- **Projection:** Final columns full_name, student_code, and grade are returned.